

Shaaktiram Balakumar

(201) 699-6404 | sbalakumar@ucsd.edu

NanoEngineering student at UCSD with a strong foundation in materials science and scientific computing, seeking to leverage technical expertise in chemical systems and data analysis to support next-generation battery and solar innovations.

EDUCATION

University of California, San Diego

San Diego, CA

Bachelor of Science in NanoEngineering

Expected 2028

- **GPA:** 3.9 / 4.0; ACT: 35
- Prospective Minor in Entrepreneurship & Innovation

PROFESSIONAL EXPERIENCE

Graphene-Quantum Dot Hybrid Photodetector

San Diego, CA

NANO 4

Fall 2025

- Developed high-performance hybrid photodetectors by integrating monolayer graphene with PbS quantum dots to maximize light absorption via the photogating effect.
 - **Nanofabrication & Thin Films:** Fabricated devices using CVD graphene wet transfer, chemical etching, and spin-coating of 15–20nm thin films.
 - **Performance Optimization:** Optimized charge transfer efficiency and validated device performance through rigorous I/V sweep analysis and photocurrent measurements.
 - **Technical Documentation:** Documented the full fabrication and testing lifecycle in a professional research report utilizing IEEE-style formatting.

BMES (Biomedical Engineering Society)

San Diego, CA

Medical Innovation Conference Sublead

August 2025 – Present

- Coordinating large-scale logistics and professional onboarding for the 2026 Medical Innovation Conference to connect 150+ students with industry leaders.
 - **Industry Outreach:** Spearheaded outreach to founders and executives, successfully securing high-level speakers from Hydrostasis, Molecular Assemblies, and Air Surgical.
 - **Operations Management:** Managed administrative agreements and start-up speaker onboarding to ensure seamless execution of the Startups Panel.

PROJECTS

Pineapple Smart Glasses

Dublin, CA

Project Lead and Head of Hardware Engineering

Spring 2025

- Led the development of AI-powered wearable hardware capable of real-time language translation with natural tone and inflection replication.
 - **Hardware Prototyping:** Engineered a functional prototype using an ESP32 microcontroller, integrating a haptic buzzer, OLED screen, and Wi-Fi modules.
 - **MVP Success:** Directed the technical showcase and hardware pitch that resulted in winning the local Entrepreneurship competition.

GaelPager [Non-Profit]

Dublin, CA

Head Engineer & Project Lead

June 2022 – August 2024

- Conceptualized and built a custom paging device to streamline communication and elevate the quality of VEX robotics tournament operations.
 - **Product Development:** Led a team of 6 to manage the full product lifecycle, from conceptualization to building and final showcase.
 - **Technical Integration:** Developed and applied advanced PCB design and programming skills to create a scalable hardware solution for tournament efficiency.

SKILLS, ACTIVITIES & INTERESTS

Languages: Fluent in English and Tamil; Conversational Proficiency in French

Technical Skills: Python, C, C++, MATLAB

Certifications & Training: 3rd Place Winner in Flux.ai National PCB Design Competition; Python Certiport Certification; American Red Cross Adult and Pediatric First Aid/CPR/AED Certified

Lab Skills: Graphene transfer, Quantum Dot Spin coating, Chromatography, Gel electrophoresis, PCR, ELISA, Cell Seeding/Propagation, Histology, and biochemical assay interpretation

Hardware & Engineering Design Skills: PCB Design (3 years), Mechanical Robotics Design (5 years), ESP32 Development, CVD Graphene Wet Transfer, and Chemical Etching